The Artificial Neuron

Name _____

Work with a partner. Compare answers when you finish each problem.

Word bank: adds compares Fighter multiplies Wizard

You can use a word more than once.

1. Fighter or Wizard? These creatures are made up.

Krellis: Strength 4, Magic 2. It's a _____. Lullom: Strength 2, Magic 4. It's a _____.

The rule I used: _____

2. Multiply, add, compare.

An artificial neuron _____ each input by its weight, _____ them up, and _____ the total with the threshold.

Strength 5, Magic 2. Weights: Strength 1, Magic -1. Threshold: 0. More than the threshold: Fighter.

Total = $(5 \times 1) + (2 \times -1)$ = _____ Fighter or Wizard? _____

Strength 2, Magic 3. Weights: Strength 2, Magic -1. Threshold: 2.

Total = _____ Fighter or Wizard? _____

3. Find the weights. Write every try. Change one number at a time.

Try	Strength weight	Magic weight	Threshold	Cards right (of 6)
Start	1	1	5	
2				
3				
4				
5				

Our weights that got 6 of 6: Strength _____ Magic _____ Threshold _____

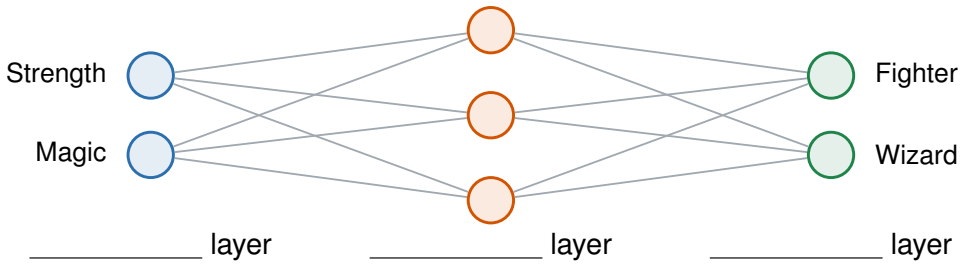
Card	Strength	Magic	Total	Neuron says	Really
1 Krellis	4	2			Fighter
2 Lullom	2	4			Wizard
3 Rezivil	5	2			Fighter
4 Pona	1	5			Wizard
5 Sandam	3	2			Fighter
6 Skabil	3	4			Wizard

Word bank: hidden input output threshold weight
You can use a word more than once.

4. Real neuron, artificial neuron. Match each part.

Real neuron	Artificial neuron
Dendrites take signals in	
Strong or weak synapses	
Fires only if the total is big enough	

5. From one neuron to a network. Name each layer.



Each line is a _____. How many lines? $2 \times 3 + 3 \times 2 =$ _____

Why is the middle layer called hidden? _____

6. Talk about it.

We fixed the weights by _____

I think an AI with billions of weights gets them by _____

Go Further (Day 2). Test your neuron on all 140 Fighters and Wizards.

Our class's neuron: _____ % right. My best: _____ % with weights _____ _____ _____

Can any line get all 140 right? _____ Why? _____